

MTH203: Sequences and series
Lecture 16

1. SUMMATION BY PARTS

The following algebraic trick will allow us to prove a generalization of the Alternating Series Test.

Theorem 1 (Summation by parts). *Let $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$ be two sequences of real numbers. For any $n \in \mathbb{N}$, we define $A_n := \sum_{k=1}^n a_k$. We also define $A_0 = 0$. Then if p and q are integers such that $1 \leq p \leq q$, we have*

$$\sum_{n=p}^q a_n b_n = \left(\sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) \right) + A_q b_q - A_{p-1} b_p.$$

Proof. For any integer, $n \in \mathbb{N}$, we have $a_n = A_n - A_{n-1}$. So,

$$\begin{aligned} \sum_{n=p}^q a_n b_n &= \sum_{n=p}^q (A_n - A_{n-1}) b_n \\ &= \left(\sum_{n=p}^q A_n b_n \right) - \left(\sum_{n=p}^q A_{n-1} b_n \right) \\ &= \left(\sum_{n=p}^q A_n b_n \right) - \left(\sum_{n=p-1}^{q-1} A_n b_{n+1} \right) \\ &= \left(\sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) \right) + A_q b_q - A_{p-1} b_p. \end{aligned}$$

□

Remark 2. The above identity should seem very similar to the “integration by parts” formula that is used in integral calculus. It is essentially the same idea.

Theorem 3. *Let $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$ be sequences of real numbers with the following properties:*

- *The sequence $(b_n)_{n \in \mathbb{N}}$ is non-increasing. In other words we have $b_n \geq b_{n+1}$ for all $n \in \mathbb{N}$.*
- *$\lim_{n \rightarrow \infty} b_n = 0$.*
- *For any $n \in \mathbb{N}$, let $A_n = \sum_{k=1}^n a_k$. Then, the sequence $(A_n)_{n \in \mathbb{N}}$ is bounded.*

Then, the series $\sum_{n=1}^{\infty} a_n b_n$ converges.

Proof. Since the sequence $(A_n)_{n \in \mathbb{N}}$ is bounded, there exists a real number M such that $|A_n| \leq M$ for all $n \in \mathbb{N}$.

Let $\epsilon > 0$ be given. By Cauchy’s Criterion for convergence of series, it suffices to prove that there exists an integer N such that for $N < p \leq q$, we have the inequality $\left| \sum_{n=p}^q a_n b_n \right| < \epsilon$.

We use the Summation by Parts formula to get the following estimate:

$$\begin{aligned}
\left| \sum_{n=p}^q a_n b_n \right| &= \left| \left(\sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) \right) + A_q b_q - A_{p-1} b_p \right| \\
&\leq \left(\sum_{n=p}^{q-1} |A_n| \cdot |b_n - b_{n+1}| \right) + |A_q b_q| + |A_{p-1} b_p| \\
&\leq \left(\sum_{n=p}^{q-1} M \cdot |b_n - b_{n+1}| \right) + M|b_q| + M|b_p| \\
&= M \cdot \left(\left(\sum_{n=p}^{q-1} |b_n - b_{n+1}| \right) + |b_q| + |b_p| \right)
\end{aligned}$$

For all $n \in \mathbb{N}$, we have $b_n \geq b_{n+1}$ for all n , and so $|b_n - b_{n+1}| = b_n - b_{n+1}$. Also, both b_p and b_q are non-negative and so $|b_p| = b_p$ and $|b_q| = b_q$. Thus, the above inequality can be rewritten as

$$\left| \sum_{n=p}^q a_n b_n \right| \leq M \cdot \left(\left(\sum_{n=p}^{q-1} b_n - b_{n+1} \right) + b_q + b_p \right) \quad (1)$$

The expression inside the innermost pair of parentheses on the right hand side can be rewritten as follows:

$$\begin{aligned}
\left[\sum_{n=p}^{q-1} b_n - b_{n+1} \right] &= [(b_p - b_{p+1}) + (b_{p+1} - b_{p+2}) + \cdots + (b_{q-1} - b_q)] \\
&= b_p - b_q
\end{aligned}$$

Substituting this in inequality (1), we get

$$\left| \sum_{n=p}^q a_n b_n \right| \leq M \cdot ((b_p - b_q) + b_q + b_p) = 2M b_p. \quad (2)$$

Since $\lim_{n \rightarrow \infty} b_n = 0$, there exists an integer N such that for all $n > N$, we have $|b_n| < \epsilon/(2M)$. Then, if we have integers p and q such that $N < p \leq q$, inequality (2) gives us

$$\left| \sum_{n=p}^q a_n b_n \right| \leq 2M b_p < 2M \cdot \epsilon/(2M) = \epsilon.$$

□

Example 4. Let $(b_n)_{n \in \mathbb{N}}$ be a sequence of real numbers satisfying the conditions given in the statement of the above theorem, and let $(a_n)_{n \in \mathbb{N}}$ be defined by $a_n = (-1)^{n+1}$ for all n . Then $(A_n)_{n \in \mathbb{N}}$ (again, defined as in the statement of the theorem) is just the sequence $1, 0, 1, 0, \dots$, which is clearly bounded. Thus, the theorem tells us that the series $\sum_{n=1}^{\infty} (-1)^{n+1} b_n$ converges. Thus, the above theorem is a generalization of the Alternating Series Test.

2. REARRANGEMENT OF SERIES

Definition 5. Let $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$ be sequences of real numbers. We say that the series $\sum_{n=1}^{\infty} b_n$ is a *rearrangement* of the series $\sum_{n=1}^{\infty} a_n$ if there exists a bijection $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ such that $b_n = a_{\sigma(n)}$ for every n .

If we have a finite collection of numbers, it does not matter what order they are added in. For example $(1 + 2) + 3 = (3 + 2) + 1$. So, it is tempting to believe that two series are rearrangements of each other, then one of them converges if and only if the other does. Also, one might believe, that if they are both convergent, they should converge to the same real number. However, this is not true.

Indeed, recall that the “infinite sum” $\sum_{n=1}^{\infty} a_n$ is supposed to be the limit of the sequence $(p_n)_{n \in \mathbb{N}}$, where $p_n = \sum_{k=1}^n a_k$. Similarly, the “infinite sum” $\sum_{n=1}^{\infty} b_n$ is supposed to be the limit of the sequence $(q_n)_{n \in \mathbb{N}}$, where $q_n = \sum_{k=1}^n b_k$. Even if the two series are rearrangements of each other, the sequences $(p_n)_{n \in \mathbb{N}}$ and $(q_n)_{n \in \mathbb{N}}$ are generally very different. Thus, a priori, it is not clear whether the convergence of one has any implications about the convergence of the other. In fact, let us look at examples where things go very wrong.

Example 6. Consider the series $\sum_{n=1}^{\infty} (-1)^{n+1}/n$. We have seen (via the Alternating Series Test) that this series converges. Let x be any real number. We will show that there exists a rearrangement of this series that converges to x .

First, we take all the numbers in the series that are positive and arrange them in decreasing order:

$$1 > 1/3 > 1/5 > \dots > 1/(2k+1) > \dots$$

Thus, the k -th number in this sequence is $1/(2k-1)$. We call this the Positive sequence.

We take all the negative numbers and arrange them in increasing order:

$$-1/2 < -1/4 < -1/6 < \dots < -1/(2k) < \dots$$

Thus, the k -th number in this sequence is $-1/(2k)$. We call this the Negative sequence.

Now, we create a new sequence by using all the numbers in both, the Positive and Negative sequences, such that the corresponding series converges to x . We will denote the n -th partial sum in this new sequence by s_n .

First, we pick some numbers from the Positive sequence in decreasing order. Since the series $\sum_{k=1}^{\infty} \frac{1}{2k-1}$ diverges, there exists some number N such that $\sum_{k=1}^N \frac{1}{2k-1} > x$. Let n_1 be the *smallest* such number, so that $\sum_{k=1}^{n_1} \frac{1}{2k-1} > x$, but $\sum_{k=1}^{n_1-1} \frac{1}{2k-1} \leq x$. Thus, $s_{n_1} = \sum_{k=1}^{n_1} \frac{1}{2k-1}$. Note that $n_1 \geq 1$. Also, observe that $1/(2n_1-1) > s_{n_1} - x \geq 0$.

Now, we pick numbers from the Negative sequence in increasing order, until the first instance where the partial sum becomes strictly less than x . Suppose this happens for the first time after picking the m_1 -th number in the Negative sequence. Thus, the partially picked series, so far, is as follows:

$$s_{(n_1+m_1)} := \frac{1}{1} + \dots + \frac{1}{2n_1-1} - \frac{1}{2} - \dots - \frac{1}{2m_1} \leq x$$

Note that $m_1 \geq 1$. Also, observe that $1/(2m_1) > x - s_{(n_1+m_1)} \geq 0$.

Now, once again we pick numbers from the Positive sequence in decreasing order, until the first instance where the partial sum becomes strictly greater than x . Suppose this happens after picking the n_2 -th number in the Positive sequence. Thus, the series so far is as follows:

$$s_{(n_2+m_1)} := \frac{1}{1} + \dots + \frac{1}{2n_2-1} - \frac{1}{2} - \dots - \frac{1}{2m_1} + \frac{1}{2n_1+1} + \dots + \frac{1}{2n_2-1}$$

Note that $n_2 > n_1$ and so $n_2 \geq 2$. Also, observe that $1/(2n_2-1) \geq s_{(n_2+m_1)} - x > 0$

We continue in this manner, causing the partial sums to sway past x one way and the other. Thus, we end up obtaining sequences integers $n_1 < n_2 < \dots$ and $m_1 < m_2 < \dots$ so that the first n_1 terms are positive, the next m_1 terms are negative, the next n_2 terms are positive, and so on. Thus, $n_i \geq i$ and $m_i \geq i$ for all i .

Thus, the terms where the partial sums “change direction” are the ones numbered $n_1, (n_1 + m_1), (m_1 + n_2), (n_2 + m_2)$, etc. One can easily verify (Exercise!) that if $n_i + m_i \leq k < n_{i+1} + m_i$, then

$$|s_k - x| \leq \max \left\{ \frac{1}{2m_i}, \frac{1}{2n_{i+1} - 1} \right\} \leq \frac{1}{2i + 1}$$

and if $n_{i+1} + m_i \leq k < n_{i+1} + m_{i+1}$, then

$$|s_k - x| \leq \max \left\{ \frac{1}{2n_{i+1} - 1}, \frac{1}{2m_{i+1}} \right\} \leq \frac{1}{2i + 1}.$$

It is easy to conclude from this that the series converges to x .

The reason for this kind of behaviour is that the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ is *conditionally convergent*. We will show that this sort of thing never happens for an *absolutely* convergent series. First, we note a rather simple lemma about convergence of sequences:

Lemma 7. *Let $(x_n)_{n \in \mathbb{N}}$ be a convergent sequence of real numbers, and let $x = \lim_{n \rightarrow \infty} x_n$. Let $(y_n)_{n \in \mathbb{N}}$. Suppose that for any $\epsilon > 0$, there exists an integer N such that for all $n > N$, we have $|x_n - y_n| < \epsilon$. Then $\lim_{n \rightarrow \infty} y_n = x$.*

Proof. Let $\epsilon > 0$ be given. There exists an integer N_1 such that for all $n > N_1$, $|x - x_n| < \epsilon/2$. Also, there exists an integer N_2 such that for all $n > N_2$, $|x_n - y_n| < \epsilon/2$.

Let $N = \max\{N_1, N_2\}$. Then, for any $n > N$, we have

$$|x - y_n| \leq |x - x_n| + |x_n - y_n| < \epsilon/2 + \epsilon/2 = \epsilon.$$

Thus, $\lim_{n \rightarrow \infty} y_n = x$. □

Exercise 8. Prove the converse of Lemma 7. In other words, prove that if $(x_n)_{n \in \mathbb{N}}$ and $(y_n)_{n \in \mathbb{N}}$ are two sequences of real numbers that converge to the same point, then for any $\epsilon > 0$, there exists N such that for all $n > N$, we have $|x_n - y_n| < \epsilon$.

We will also need the following lemma about absolutely convergent series:

Lemma 9. *Let $\sum_{n=1}^{\infty} x_n$ be an absolutely convergent series of real numbers. Then, for any $\epsilon > 0$, there exists an integer N such that if $A \subset \mathbb{N}$ is any finite subset such that $A \cap \{1, \dots, N\} = \emptyset$, then $|\sum_{k \in A} x_k| < \epsilon$.*

Proof. Let $\epsilon > 0$ be given. Since $\sum_{n=1}^{\infty} |x_n|$ is a convergent series, there exists an integer N such that for any integer $n \geq m > N$, $\sum_{k=m}^n |x_k| < \epsilon$.

Let $m = \min(A)$ and let $n = \max(A)$. Then, as $A \cap \{1, \dots, N\} = \emptyset$, we have $N < m \leq n$. Thus,

$$\left| \sum_{k \in A} x_k \right| \leq \sum_{k \in A} |x_k| \leq \sum_{k=m}^n |x_k| < \epsilon.$$

□

Exercise 10. Prove that the conclusion of Lemma 9 fails for conditionally convergent sequences. (For example, prove that it is not true for $\sum_{n=1}^{\infty} (-1)^{n+1}/n$.)

However, we have the following important result:

Theorem 11. *Let $\sum_{n=1}^{\infty} a_n$ be an absolutely convergent series of real numbers. Then, any rearrangement of the series converges to $\sum_{n=1}^{\infty} a_n$.*

Proof. Now, let $\sum_{n=1}^{\infty} b_n$ be a rearrangement of $\sum_{n=1}^{\infty} a_n$. Thus, there is a bijection $\sigma : \mathbb{N} \rightarrow \mathbb{N}$ such that $b_n = a_{\sigma(n)}$. Let $\tau = \sigma^{-1}$, so that $b_{\tau(n)} = a_n$ for all n .

For any integer n , let $s_n = \sum_{k=1}^n a_k$ and $t_n = \sum_{k=1}^n b_k$. By Lemma 7, to show that the sequences $(s_n)_{n \in \mathbb{N}}$ and $(t_n)_{n \in \mathbb{N}}$ converge to the same point, it suffices to

show that for any $\epsilon > 0$, there exists an integer M such that for any $n > M$, we have $|s_n - t_n| < \epsilon$.

Let $\epsilon > 0$ be given. The hypothesis says that the series $\sum_{n=1}^{\infty} |a_n|$ converges. Thus, by Lemma 9, for any $\epsilon > 0$, there exists an integer N such that for any finite subset $A \subset \mathbb{N}$ such that $A \cap \{1, \dots, N\} = \emptyset$, we have $\sum_{k \in A} |a_k| < \epsilon/2$. Let

$$M := \max\{\tau(1), \dots, \tau(N)\}.$$

Note that M is greater than or equal to the N distinct positive integers $\sigma(1), \dots, \sigma(N)$, and so $M \geq N$.

Let $n > M$. Let $S = \{\tau(1), \dots, \tau(N)\}$ and let $T = \{1, \dots, n\} \setminus S$. Then we have

$$\begin{aligned} t_n &= \sum_{k=1}^n b_k = \left(\sum_{k \in S} b_k \right) + \left(\sum_{k \in T} b_k \right) \\ &= \left(\sum_{k=1}^N b_{\sigma(k)} \right) + \left(\sum_{k \in T} b_k \right) = \left(\sum_{k=1}^N a_k \right) + \left(\sum_{k \in T} b_k \right). \end{aligned}$$

We also have

$$s_n = \left(\sum_{k=1}^N a_k \right) + \left(\sum_{k=N+1}^n a_k \right).$$

Thus,

$$t_n - s_n = \left(\sum_{k \in T} b_k \right) - \left(\sum_{k=N+1}^n a_k \right)$$

and so

$$|t_n - s_n| \leq \left| \sum_{k \in T} b_k \right| + \left| \sum_{k=N+1}^n a_k \right|.$$

By the choice of N , we have

$$\left| \sum_{k=N+1}^n a_k \right| < \epsilon/2.$$

To obtain an estimate for the expression $|\sum_{k \in T} b_k|$, we note that if $k \in T$, then $k = \tau(n)$ for some $n > N$. In other words, $\sigma(k) > N$. Thus, $\sigma(T) \cap \{1, \dots, N\} = \emptyset$. So,

$$\begin{aligned} \left| \sum_{k \in T} b_k \right| &= \left| \sum_{n \in \sigma(T)} b_{\tau(n)} \right| \\ &= \left| \sum_{n \in \sigma(T)} a_n \right| < \epsilon/2. \end{aligned}$$

Using the last two inequalities, we get

$$|t_n - s_n| < \epsilon/2 + \epsilon/2 = \epsilon.$$

This proves the result. $\square$

3. READING SUGGESTIONS

The results in Section 1 is from [2, Theorems 3.41 and 3.42]. The rest of the material is from Sections 2.1 and 2.7 of the textbook ([1]).

REFERENCES

- [1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015. [5](#)
- [2] Walter Rudin, *Principles of Mathematical Analysis*. 3rd ed., McGraw-Hill, 1976. [5](#)